REASEARCH PAPER

AI-Based Flood Monitoring and Early Warning System for the Beas River Basin with Real-World Case Analysis

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Abstract->

Floods are among the most destructive natural disasters, especially in river basins like the Beas River, where sudden rainfall, dam discharge, and glacial melt create high flood risk. This paper presents an **AI-based flood monitoring and early warning system** integrating IoT sensors and machine learning models.

The study incorporates **real-world flood scenarios**, including the **2018 Beas River flood incident**, where sudden water release from Larji Dam caused casualties among tourists. The proposed system uses models such as Random Forest and LSTM to predict flood risks using real-time environmental data.

The results demonstrate that AI-based systems can provide early warnings and significantly reduce disaster impact.

Keywords->

Flood Prediction, Artificial Intelligence, Beas River, IoT, LSTM, Random Forest, Disaster Management, Early Warning System

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1. Introduction->

Floods are recurring natural disasters that cause large-scale destruction globally. In India, rivers originating from the Himalayas, including the Beas River, are highly flood-prone due to:

- Intense monsoon rainfall
- Glacier melting
- Sudden dam water release

A tragic example occurred in **June 2014**, when sudden water release in the Beas River led to the death of **24 engineering students** near Thalout. This incident highlights the urgent need for **real-time monitoring and early warning systems**.

Traditional systems fail to provide timely alerts, whereas AI-based systems can analyze real-time data and predict flood risks efficiently.

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2. Literature Review with Real Applications->

Several real-world implementations support AI-based flood prediction:

- During the Kerala Floods 2018, lack of coordinated data analysis worsened the disaster. AI could have improved prediction accuracy.
- AI-based flood prediction models in China's Yangtze River basin have successfully predicted floods using LSTM models.
- IoT-based systems in Europe provide real-time river monitoring using sensor networks.

These examples prove that integrating AI with real-time data significantly improves flood prediction and management.

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3. Problem Statement->

The Beas River basin faces frequent flood risks due to:

- Sudden release of water from dams like Larji Dam
- Heavy rainfall in upstream regions
- Lack of automated alert systems

Real Situation Example

In 2014:

- Water was released without proper warning
- Tourists near the river were unaware
- No automated alert system existed

This resulted in **avoidable loss of life**, highlighting the need for an intelligent system.

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4. Proposed System with Real-Life Mapping->

The proposed system integrates **AI + IoT + Cloud Computing**.

4.1 Real-Time Scenario Working

Imagine:

- Heavy rainfall occurs in Himachal Pradesh
- Water level rises rapidly in the Beas River
- Sensors detect abnormal increase
- AI model predicts flood risk within minutes
- Alerts are sent to nearby villages and tourists

This could have prevented incidents like the 2014 tragedy.

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5. Methodology with Practical Example->

5.1 Data Collection

- Historical flood data of Beas River
- Rainfall data from meteorological departments

Example:
During monsoon, rainfall exceeds threshold → system records spike

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5.2 Data Preprocessing

- Remove missing data
- Normalize values

Example:
Sensor error corrected automatically before prediction

5.3 Model Training

- Train using past flood events

Example:
Model learns pattern:

- High rainfall + rising water level = flood risk

5.4 Prediction Process

- Real-time data input
- AI predicts flood probability

Example Output:

- Flood Risk = 85% → Alert triggered

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6. System Architecture->

Sensors → Microcontroller → Cloud → AI Model → Alert System

Real-Life Flow Example

1. Rainfall sensor detects heavy rain
2. Water level sensor detects rapid increase
3. Data sent to cloud
4. AI predicts flood
5. SMS alert sent to residents

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7. Results and Discussion with Case Insight->

Case Study: 2014 Beas River Incident

- Cause: Sudden dam water release
- Problem: No warning system
- Result: Casualties

With Proposed System

- Sensors detect rising water
- AI predicts surge
- Alerts sent instantly
- People evacuated

Outcome: Lives saved

Case Study: Kerala Floods 2018

- Lack of real-time coordination
- AI could have improved prediction

AI systems today can:

- Predict floods hours/days earlier
- Provide location-specific alerts

8. Advantages (with Practical Impact)->

Feature	Real Impact
Early warning	Saves lives (tourists, villagers)
Real-time monitoring	Prevents sudden disasters
Automation	Reduces human error
AI prediction	Improves accuracy

9. Limitations->

- Requires proper sensor installation
- Internet connectivity issues in remote areas
- Initial setup cost

10. Future Scope->

- Integration with satellite data (ISRO)
- Drone-based flood monitoring
- Mobile apps for tourists near rivers
- Smart dam control systems

11. Conclusion->

The AI-based flood monitoring system for the Beas River provides a **practical and life-saving solution** to flood disasters. By learning from real incidents such as the 2014 Beas River tragedy and the Kerala floods, the proposed system demonstrates how **technology can prevent loss of life and property**.

The integration of AI, IoT, and real-time data analytics makes this system highly effective for modern disaster management.

12. References->

1. Government reports on Beas River flood incidents
2. Studies on Kerala Floods 2018
3. Research papers on AI-based flood prediction
4. IoT-based environmental monitoring studies